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Lagranges Interpolation

$l_i(X_j) = \delta_{ij}$

corresponding to x_0, we have polynomial

$l_0(X_0) = 1$	$l_1(X_0) = 0$	$l_n(X_0) = 0$
$l_0(X_1) = 0$	$l_1(X_1) = 1$	$l_n(X_1) = 0$
$l_0(X_2) = 0$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
$l_0(X_n) = 0$	$l_1(X_n) = 0$	$l_n(X_n) = 1$

$$l(X) = \sum_{i=0}^{n} y_i l_i(X) \rightarrow \text{Lagrange's interpolation}$$

$$l_0(X_0) = (X - X_1)(X - X_2) \cdots (X - X_n)$$

$$l_0(X) = \frac{(X - X_1)(X - X_2) \cdots (X - X_n)}{(X_0 - X_1)(X_0 - X_2) \cdots (X - X_n)}$$

$$l_1(X) = \frac{(X - X_0)(X - X_2) \cdots (X - X_n)}{(X_1 - X_0)(X_1 - X_2) \cdots (X_1 - X_n)}$$

$$l_i(X) = \frac{(X - X_0) \cdots (X - X_{i-1})(X - X_{i+1}) \cdots (X - X_n)}{(X_i - X_0) \cdots (X_i - X_{i-1})(X_i - X_{i+1}) \cdots (X_i - X_n)}$$

1	2	3	4
0	1/3	2/3	1

Construct a interpolation polynomial passing from (0,1), (1/3, 2), (2/3, 3), (1,4)

$$l_0(x) = \frac{(X-1/3)(X-2/3)(X-1)}{(0-\frac{1}{3})(0-2/3)(0-1)}$$

$$l_1(x) = \frac{(X-0)(X-2/3)(X-1)}{(\frac{1}{3}-0)(\frac{1}{3}-2/3)(\frac{1}{3}-1)}$$

$$l_2(x) = \frac{(X-0)(X-\frac{1}{3})(X-1)}{(\frac{2}{3}-0)(\frac{2}{3}-\frac{1}{3})(\frac{2}{3}-1)}$$

$$l_3(x) = \frac{(x-0)(x-\frac{1}{3})(x-2/3)}{(1-0)(1-\frac{1}{3})(1-\frac{2}{3})}$$

Newton's Interpolation :-

y_0	y_1			y_n
x_0	x_1	x_2	— — —	x_n

$$p_0(x) - y_0 = \frac{y_1 - y_0}{x_1 - x_0}(x - x_0)$$

$\cancel{p_1(x)}$

$$p_1(x) = y_0 + \frac{y_1 - y_0}{(x_1 - x_0)}(x - x_0)$$

$$p_2(x) = y_0 + \frac{(y_1 - y_0)}{(x_1 - x_0)}(x - x_0)$$

$$y_2(x) = y_0 + \frac{(y_1 - y_0)}{(x_1 - x_0)}(x_2 - x_0)$$

$$+ A_2(x_2 - x_0)(x_2 - x_1)$$

$$\cancel{y_2 - y_0 = \frac{y_2 - p_1(y_2)}{x - x}}$$

$$A_2 = \frac{y_2 - p_1(y_2)}{(x_2 - x_0)(x_2 - x_1)}$$

$$A_n = \frac{\cancel{y_n - p_{n-1}(x_n)} \; y_n - p_{n-1}(x_n)}{(x_n - x_0)(x_n - x_1) \cdots (x_n - x_{n-1})}$$

$$p_1(x_2) = y_0 + \frac{(y_1 - y_0)(x_2 - x_0)}{(x_1 - x_0)}$$

$$p_n(x) = A_0 + A_1(x - x_0) + A_2(x - x_0)(x - x_1) + \cdots + A_{n+1}(x - x_0)(x - x_1) \cdots (x - x_n)$$

Ans

$$A_i = \frac{y_i - p_{i-1}(x_i)}{(x_i - x_0) \cdots (x_i - x_{i-1})}$$

$$A_1 = \frac{y_1 - y_0}{x_1 - x_0}$$

$$A_2 = \frac{y_2 - p_1(x_2)}{(x_2 - x_0)(x_2 - x_1)}$$

$$A_2 = \frac{y_2 - y_0 - \frac{(y_1 - y_0)}{(x_1 - x_0)}(x_2 - x_0)}{}$$

$$= \frac{(y_2 - y_0)(x_2 - x_0) - (y_1 - y_0)(x_2 - x_0)}{(x_1 - x_0)(x_2 - x_0)(x_2 - x_0)}$$